

Self-Supervised MKA-CNN: A Label-Efficient Multi-Kernel Attention Framework for Multimodal Gesture Decoding

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Abstract—The precise recognition of hand gestures using simultaneous surface electromyography (sEMG) and accelerometry (ACC) signals is a critical frontier in the development of intuitive human–computer interfaces and advanced neuroprosthetics. Current state-of-the-art deep learning architectures remain heavily bottlenecked by their reliance on massive, meticulously annotated datasets—a requirement that is both clinically prohibitive and operationally impractical in real-world medical settings. This study rigorously evaluates the capacity of Self-Supervised Learning (SSL) to bridge this fundamental “label gap”. We present a comprehensive comparative analysis of three distinct SSL frameworks—SimCLR, BYOL, and Temporal Neighborhood Coding (TNC)—applied to multimodal physiological data from the NinaPro DB7 dataset. Proposing Multi Kernel Attention 1D Convolutional Neural Network (CNN) backbone as the encoder for all frameworks, we systematically benchmarked these SSL approaches against a fully supervised Multi Kernel Attention CNN and a classical Deep Neural Network (DNN) baseline. Our results demonstrate that SSL pre-training, particularly via BYOL, achieves classification accuracies up to 98.92%—close to fully supervised methods—without any human-labeled data during pre-training. Analysis of training dynamics further reveals distinct convergence profiles for each SSL paradigm, providing mechanistic insight into how each framework captures the latent structure of myoelectric signals. This work underscores the transformative potential of self-supervised representation learning in overcoming the data-dependency constraints of modern biomedical signal processing.

Index Terms—Surface Electromyography, Accelerometry, Self-Supervised Learning, Multi-Kernel Attention, SimCLR, BYOL, Temporal Neighborhood Coding, Convolutional Neural Network, NinaPro DB7, Gesture Recognition, Neuroprosthetics.

I. Introduction

The accurate decoding of neuromuscular intent via surface electromyography (sEMG) constitutes one of the most active and consequential research frontiers in biomedical engineering. By capturing the quality of electrical activity of motor units firing beneath the skin surface, sEMG provides a non-invasive, high-temporal-resolution window into the human peripheral nervous system [1]. This capability is crucial for a diverse range of applications: next-generation prosthetic limbs that move with natural, intuitive control; rehabilitation systems that monitor and guide motor recovery; and gesture-based human–computer interfaces that enable hands-free device operation [2], [5].

Despite decades of progress, the field faces a persistent and deeply frustrating bottleneck: the chronic scarcity of well-annotated training data. The stochastic realities of sEMG signals—characterized by substantial noise arising from physiological sources, sensitivity to electrode placement and skin impedance, and profound inter- and intra-subject variability due to differences in anatomy, fatigue, and session-to-session electrode repositioning—make the problem of building universally generalizable supervised models enormously challenging [3]. Furthermore, the process of collecting labeled sEMG data is itself clinically expensive: it demands the sustained, fatiguing effort of trained subjects executing specific, precisely-timed movements under structured laboratory conditions. Every millisecond of signal must be painstakingly matched to a ground-truth gesture label.

Self-Supervised Learning (SSL) offers a compelling theoretical answer to this labeling bottleneck. Rather than relying on costly human annotations, SSL frameworks employ carefully designed pretext tasks to autonomously discover the intrinsic statistical and structural regularities of raw, unlabeled data. The learned representations can then be efficiently transferred to downstream classification tasks using only a small fraction of labeled examples. Seminal SSL frameworks such as SimCLR

[6], which operates by maximizing agreement between two differently augmented views of the same input via contrastive loss, and BYOL [7], which eliminates negative pairs through an asymmetric bootstrapping mechanism, have demonstrated near-supervised-level performance on visual benchmarks. TNC [8], designed with the temporal structure of physiological time-series in mind, leverages the continuity property of muscle activation patterns to define a principled self-supervised objective.

To the best of our knowledge, this is the first study to conduct a controlled, head-to-head evaluation of SimCLR, BYOL, and TNC within a single unified experimental framework on the NinaPro DB7 multimodal sEMG dataset. By controlling for backbone architecture and using identical preprocessing and evaluation protocols, we isolate the contribution of each SSL paradigm to the learned representation quality. This work makes the following key contributions:

- 1) We present the a systematic comparison of three SSL frameworks (SimCLR, BYOL, TNC) on multimodal sEMG-ACC data using a shared Multi-Kernel Attention 1D CNN backbone;
- 2) We demonstrate that SSL pre-training achieves accuracy within 0.20% of fully supervised methods while requiring zero labeled data during pre-training;
- 3) We provide a mechanistic analysis of training dynamics and convergence behavior unique to each SSL paradigm on physiological time-series data; and
- 4) We propose the Multi-kernel Attention(MKA) 1D CNN, a computationally efficient architecture that achieves state-of-the-art accuracy(98.72%) with 10.7x fewer parameters than traditional deep residual network, specially optimized for real-time edge inference.

II. Related Work

Recent work on sEMG-based gesture recognition has largely focused on deep learning models utilizing the high-density NinaPro DB7 dataset. The evolution of these methodologies can be categorized into standard supervised baselines, deep multimodal fusion, and emerging self-supervised foundation models.

A. Supervised Baselines and Label Refinement

Early deep learning approaches established robust supervised baselines using purely myoelectric signals. P. Sri-iesaranusorn et al. [4] utilized a deep neural network to classify 41 hand and wrist movements, achieving 91.69% accuracy. Similarly, A. Fatayer et al. [3] built a standard CNN for DB7, reporting 91.7% accuracy. Addressing the inherent stochasticity of clinical data, Y. Chen et al. [7] studied the effect of noisy labels, proposing an Multi-Kernel Attention CNN (MKA-CNN) that iteratively refines suspected mislabeled trials to yield 95.85% accuracy. While highly accurate, such methods require complex training procedures.

B. Classical Feature Engineering Approaches

The earliest and most foundational line of work in sEMG-based gesture recognition relied on the manual extraction of hand-crafted time-domain and frequency-domain features from raw signal windows, followed by classical classifiers. Common features included Root Mean Square

(RMS) energy, Mean Absolute Value (MAV), Waveform Length (WL), Zero Crossing Rate (ZCR), and slope sign changes. Sri-iesaranusorn et al. [4] demonstrated the viability of a Deep Neural Network (DNN) trained on such features for classifying 41 distinct hand and wrist movements, achieving competitive results on the NinaPro dataset family. While effective, this paradigm is inherently limited: manual feature design discards rich nonlinear spatio-temporal dynamics embedded in raw waveforms and requires domain expertise to re-engineer for each new electrode configuration or movement vocabulary.

C. Deep Multimodal Fusion

Recognizing the limitations of shallow feature engineering, subsequent research embraced end-to-end deep learning with multimodal sensor fusion. Chen et al. [1] proposed the Multi-Feature Fusion Network (MFF-Net), a multi-branch CNN architecture incorporating channel attention mechanisms that fuses complementary representations from sEMG and ACC signals, achieving 92.47% accuracy on intact subjects in NinaPro DB7. The work demonstrated that kinematic acceleration data, capturing the spatial trajectory of limb movement, provides physiologically complementary information to purely myoelectric signals.

Duan et al. [2] advanced this paradigm with their HyFusion framework, which employs parallel intramodal and intermodal processing branches, each augmented with multiscale spatial attention. By jointly modeling modality-specific features and cross-modal interactions, HyFusion established a benchmark of 96.44% accuracy on NinaPro DB7. These supervised multimodal architectures set a high performance ceiling but remain entirely dependent on complete label availability—a critical practical limitation in clinical data collection contexts. Exploring non-backpropagation methods, Y. Fang et al. [5] applied a Multi-Modality Deep Forest (MMDF) to fuse sEMG and ACC, achieving ~91.4% accuracy on intact subjects.

D. Transformers and Self-Supervised Learning (SSL)

More recently, transformer-based and pre-training approaches have emerged, suggesting SSL can dramatically improve generalization across signal variabilities. M. Fasulo et al. [13] introduced TinyMyo, a compact 3.6M-parameter transformer-encoder foundation model pre-trained via self-supervision, achieving 89.4% on DB5. Concurrently, Y. Chen et al. [14] proposed PhysioWave, a self-supervised wavelet-transformer that adaptively learns wavelet bases and selective masking to set state-of-the-art accuracy on DB5. For continuous estimation, C. Lin et al. [15] proposed a Parallel-Efficient Transformer (PET) for hand-kinematics regression, achieving 88.72% performance with extremely fast inference optimized for low-latency edge use.

E. Robustness, Label Noise, and Annotation Scarcity

A pivotal line of research has investigated the fragility of supervised deep learning models under real-world label imperfections. Fatayer et al. [3] conducted a systematic study demonstrating that deep learning classifiers for sEMG-based gesture recognition exhibit severe performance degradation when trained on noisy or inaccurate labels—a pervasive condition in real-world clinical data pipelines where expert annotators may disagree on movement boundaries or where

automated labeling systems introduce systematic biases. This finding directly motivates the pursuit of self-supervised approaches that bypass the labeling process entirely.

More recently, Fratti et al. [9] proposed a Multi-Scale CNN (MSCNN) pre-trained with self-supervision via triplet margin loss combined with adversarial domain alignment, achieving 97% F1-score across 14 movement classes on NinaPro benchmarks. Their work highlighted the dual benefit of self-supervised pre-training for both inter-subject generalization and rapid user-specific fine-tuning with minimal labeled data, demonstrating the practical clinical value of label-efficient learning pipelines.

F. Foundations of Self-Supervised Learning

The theoretical foundations of our SSL frameworks originate in the visual representation learning literature. Chen et al. [6] introduced SimCLR, demonstrating that maximizing representational agreement between two augmented views of the same sample via a contrastive NT-Xent loss, combined with a non-linear projection head, could yield representations competitive with supervised models. A critical insight was that strong data augmentation—including random cropping, color jitter, and Gaussian blur for images—is essential for learning meaningful invariances.

Grill et al. [7] subsequently introduced BYOL, which fundamentally challenged the assumption that negative samples are necessary for contrastive learning. By employing an asymmetric dual-network architecture—an online network updated by gradient descent and a slowly-evolving target network updated by exponential moving average of online weights—BYOL avoids representation collapse without requiring explicit negative pair mining, yielding improved stability under smaller batch sizes.

Tonekaboni et al. [8] extended self-supervised learning explicitly to physiological time-series with TNC, leveraging the temporal smoothness inherent in continuous biosignals. TNC defines positive samples as signal windows drawn from a temporal neighborhood of the anchor window, under the assumption that physiological states evolve continuously and adjacent windows share the same underlying neuromuscular state. This principled, domain-aware approach makes TNC uniquely suited to sEMG, where a hand gesture corresponds to a sustained, temporally correlated muscle activation sequence.

G. Self-Supervised Learning for Biosignals

Recent work has begun to extend SSL to biomedical time-series beyond the imagery domain. Kang et al. [10] proposed a unified framework coupling self-supervised temporal-spectral pre-training with adversarial domain alignment for cross-subject sEMG recognition, achieving 89.4% accuracy under leave-one-subject-out evaluation on NinaPro DB2—demonstrating that SSL pre-training can learn subject-invariant physiological representations. Fang et al. [5] demonstrated complementary modality fusion for sEMG and ACC signals using a multi-modality deep forest architecture, further validating the informational synergy between these two signal types. Our work builds on and synthesizes these advances, providing the first rigorous SSL head-to-head comparison on NinaPro DB7.

III. Methodology

A. Dataset and Multimodal Preprocessing

All experiments utilize Exercise B of the NinaPro DB7 (Non-Invasive Adaptive Prosthetics) dataset—a high-density, publicly available benchmark for hand gesture recognition collected under institutional ethics approval. NinaPro DB7 captures simultaneous sEMG and inertial measurement data from amputee and non-amputee subjects during structured, repetitive gesture trials. We focused exclusively on 17 movement classes corresponding to Classes 13–29 of Exercise B, encompassing a physiologically diverse movement vocabulary: isolated finger flexions and extensions, compound grip patterns, and dynamic wrist rotations spanning multiple degrees of freedom. We focused exclusively on Exercise B, which consists of 17 distinct hand and wrist configurations. This exercise includes 8 isometric and isotonic hand configurations—such as finger flexion and extension—and 9 basic movements of the wrist. Signals were recorded using a Delsys Trigno Wireless EMG system, which utilizes 12 active double-differential wireless electrodes sampled at 2 kHz. These sensors provide simultaneous surface electromyography (sEMG) and tri-axial accelerometry (ACC), capturing both the electrical activation of the forearm muscles and the resulting spatial kinematic trajectories.

Split Ratio	TinyMyo	DNN	DeepCNN	MKA 1D-CNN
90:10	0.9781	0.9283	0.9803	0.9872*1 1
80:20	—	0.9230	0.9756	0.9870
70:30	—	0.9113	0.9738	0.9793
60:40	—	0.8963	0.9659	0.9767
50:50	—	0.8848	0.9407	0.9550
40:60	—	0.8604	0.9536	0.9615
30:70	—	0.8288	0.9331	0.9390

20:80	—	0.7667	0.9061	0.9308
10:90	—	0.6844	0.8515	0.8875

The multimodal input stream integrates 12 high-density sEMG channels—placed circumferentially around the forearm to capture muscle activity from all major forearm compartments—and 36 accelerometry channels derived from 12 tri-axial inertial measurement units co-located with the sEMG electrodes, yielding spatial motion trajectories along all three Cartesian axes. This produces a 48-channel input that encodes a complete physiological and kinematic profile of each gesture.

Raw signals underwent standardized preprocessing: bandpass filtering of sEMG between 20 Hz and 450 Hz followed by a 50 Hz notch filter to suppress power line interference; ACC signals were low-pass filtered at 10 Hz. All channels were z-score normalized per-channel per-subject to remove inter-electrode gain differences. Signals were then segmented using a 200 ms sliding window with 50% (100 ms) overlap, a widely-validated window duration that captures the stationarity of a single gesture phase while maximizing sample diversity. This procedure yielded a total of 87,614 multivariate windows, constituting the full dataset for both unlabeled SSL pre-training and supervised benchmarking.

B. Data Augmentation Strategy

To improve the robustness of self-supervised representation learning, an asymmetric augmentation strategy was used to generate two different views of each input window. A weak view was created using Gaussian jitter and amplitude scaling, preserving the main signal structure while introducing small variations. A strong view applied a richer set of transformations, including Gaussian jitter, amplitude scaling, time shifting, time masking, channel dropping, and frequency masking. These augmentations simulate realistic variations in physiological signals, such as sensor noise, gain drift, temporal misalignment, signal dropout, electrode failure, and frequency-domain disturbances. This weak-strong design encourages the model to learn stable and invariant feature representations from multimodal sEMG and accelerometer data.

C. Multi-Kernel Attention 1D CNN Architecture

To ensure a strictly fair and controlled comparison, all three models (SimCLR, BYOL, TNC) share an identical Multi-Kernel Attention Custom 1D CNN encoder backbone. The architecture was designed to efficiently capture multi-scale temporal features from 48-channel physiological signal windows. Table II details the architectural specification.

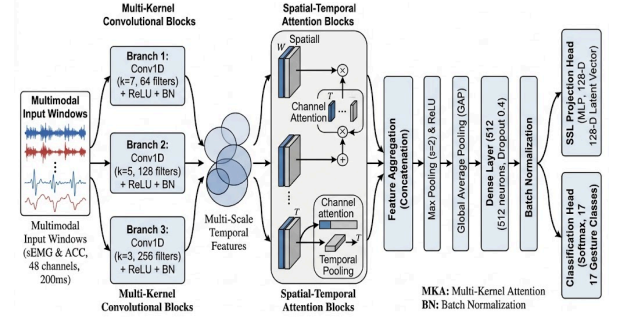


Figure 1: Proposed Multi-Kernel Attention 1D CNN Encoder Architecture.

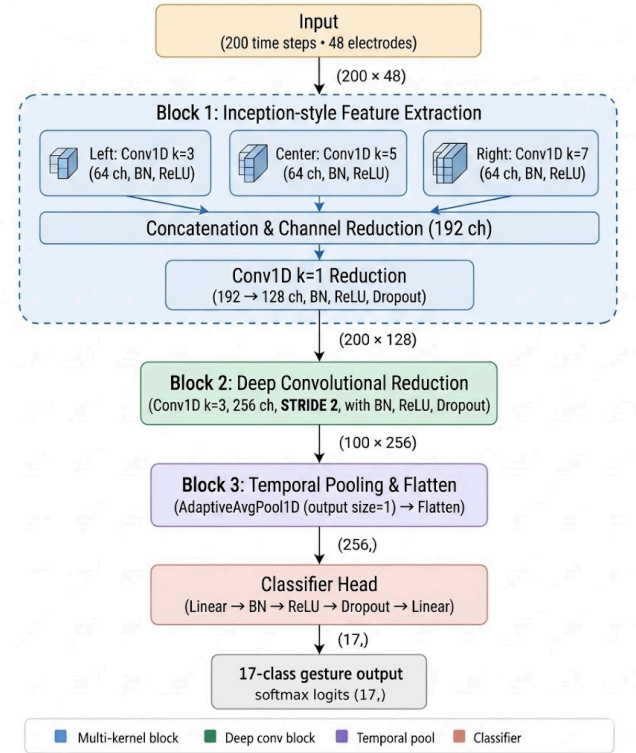


Figure 2: Structural schematic and detailed layer-wise configuration of the proposed Multi-Kernel Attention (MKA) 1D CNN encoder architecture.

TABLE I
Multi-Kernel Attention 1D CNN Backbone Architecture

Layer	Output Shape	Params
Conv1D(64, k=7) + BN + ReLU	(T, 64)	~21.5 K
MaxPool1D(2)	(T/2, 64)	—
Conv1D(128, k=5) + BN + ReLU	(T/2, 128)	~41.1 K
MaxPool1D(2)	(T/4, 128)	—
Conv1D(256, k=3) + BN + ReLU	(T/4, 256)	~98.6 K
Conv1D(256, k=3) + BN + ReLU	(T/4, 256)	~197.1 K
GlobalAvgPool1D	(256,)	—

Layer	Output Shape	Params
FC(512) + BN + ReLU	(512,)	~131.6 K
Dropout(0.4)	(512,)	—
Projection Head FC(128)	(128,)	~65.7 K

The backbone uses batch normalization (BN) after each convolutional layer to stabilize training across the diverse batch compositions inherent to SSL. A global average pooling layer collapses the temporal dimension into a fixed-length feature vector, enabling processing of variable-length windows. During SSL pre-training, a two-layer MLP projection head maps the 512-dimensional backbone representation to a 128-dimensional latent space for contrastive/bootstrapping loss computation. During fine-tuning, the projection head is discarded and replaced with a softmax classifier over 17 gesture classes.

D. Self-Supervised Learning Framework Implementations

1) SimCLR (Simple Contrastive Learning of Representations):

For each mini-batch of 256 signal windows, two stochastic augmented views are generated for every sample, resulting in 512 views in total. Each augmented view is passed through the online CNN encoder $f(\cdot)$ to obtain a feature representation h , which is then mapped to a latent vector z by the MLP projection head $g(\cdot)$. The SimCLR framework (figure 3) is trained using a contrastive objective that maximizes the similarity between the two augmented views of the same original signal window while minimizing similarity with the remaining in-batch samples, which act as negative pairs. This training strategy encourages the encoder to learn discriminative and augmentation-invariant representations useful for downstream gesture classification.

SimCLR Framework

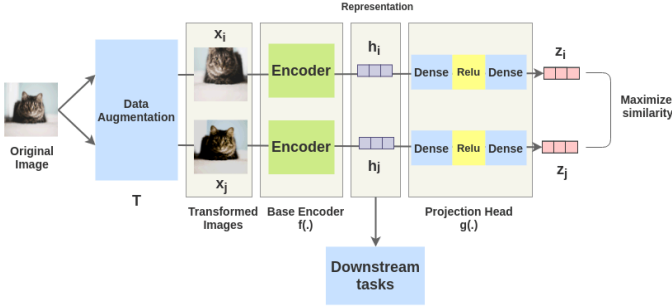


Figure 3: SimCLR architecture [16].

2) BYOL (Bootstrap Your Own Latent):

BYOL employs two network instances that share the CNN backbone architecture but are updated by different mechanisms. The online network θ (encoder f_θ + projector g_θ + predictor q_θ) is trained via gradient descent. The target network ξ (encoder f_ξ + projector g_ξ) is updated by exponential moving average: $\xi \leftarrow \tau \xi + (1-\tau)\theta$ with momentum $\tau = 0.996$. This asymmetric architecture (figure 4), combined with a predictor network present only in the online branch, prevents representational collapse without any negative examples. BYOL was trained for 250 epochs using LARS optimizer with learning rate 0.3 and cosine annealing.

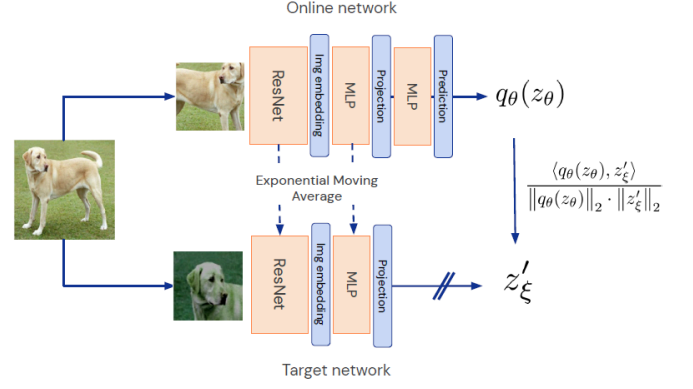


Figure 4: BYOL architecture. [Image Source](#) + annotations by [Sascha Kirch](#). [17]

3) TNC (Temporal Neighborhood Coding):

TNC (figure 5) exploits the temporal autocorrelation structure inherent in continuous physiological signals. An anchor window x_t is sampled from the signal stream. Positive (neighborhood) windows are sampled uniformly from within a temporal neighborhood $[-\Delta, +\Delta]$ of t , while negative (non-neighborhood) windows are sampled from the remainder of the recording. A discriminator $D(x_t, x_{2019})$ is trained to distinguish temporal neighbors from non-neighbors, with the CNN backbone serving as the shared feature extractor. The resulting representation captures the neuromuscular state manifold underlying continuous gesture sequences. TNC used identical CNN backbone and was trained for 250 epochs with Adam optimizer, $\alpha = 5 \times 10^{-4}$.

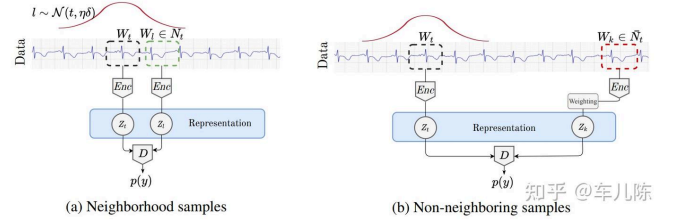


Figure 5: TNC architecture

E. Mathematical Formulations Dataset and Multimodal Preprocessing

The core optimization for SimCLR minimizes the Normalized Temperature-Scaled Cross-Entropy (NT-Xent) loss. For a positive pair (i, j) in a batch of $2N$ augmented views, the per-pair loss is:

$$\ell_i = -\log \exp(\text{sim}(z_i, z_j) / \tau)$$

$$\sum_{2' \neq i, j} \exp(\text{sim}(z_i, z_{2'}) / \tau)$$

(1)

where $\tau = 0.1$ is the temperature hyperparameter, z_i and z_j are L^2 -normalized projected representations of two augmented views of the same window, $\text{sim}(\cdot, \cdot)$ denotes cosine similarity, and the denominator sums over all $2N$ samples in the batch except the anchor i itself. The final batch loss averages ℓ_i over all positive pairs (i, j) and (j, i) .

The BYOL regression loss computes a normalized Mean Squared Error between the online prediction and the target

projection, encouraging representational consistency under augmentation without contrastive negatives: [7]

$$\mathcal{L}^{ByoL} = 2 - 2 \cdot (q\theta(z\theta) \cdot z\tilde{\zeta}) / (|q\theta(z\theta)| \cdot |z\tilde{\zeta}|) \quad (2)$$

where $q\theta$ is the online predictor and $z\tilde{\zeta}$ is the stop-gradient target projection. The TNC objective is a binary cross-entropy discriminator loss: [8]

$$\mathcal{L}^{Tnc} = E[\log D(x_{\square}, x_{\square+\delta})] + E[\log(1 - D(x_{\square}, x'))] \quad (3)$$

where $D(\cdot, \cdot)$ is a binary discriminator operating on backbone representations, $x_{\square+\delta}$ is a temporal neighbor window, and x' is a non-neighbor window sampled outside the neighborhood Δ .

F. Supervised Baselines

Four supervised baselines are employed to contextualize the performance of the self-supervised frameworks:

Proposed MKA-CNN (Multi-Kernel Attention CNN): Our primary architecture, utilizing hierarchical kernels and spatial-temporal attention. It is trained end-to-end on raw windows to represent the optimized performance ceiling for edge-deployable encoders.

DeepCNN (Deep Residual 1D CNN): A high-capacity baseline consisting of a deep 1D convolutional residual network with 5,941,649 parameters. It represents the "brute-force" deep learning approach, trading high computational cost for raw classification accuracy.

TinyMyo (Transformer-Encoder Foundation Model): A compact foundation model architecture proposed by Fasulo et al. [13]. Utilizing a transformer-encoder backbone with approximately 3.6 million parameters, it serves as a modern non-convolutional benchmark that leverages self-attention mechanisms to capture temporal dependencies in physiological signals.

DNN (Feed-Forward DNN): A classical baseline reproducing the work of Sri-iesaranusorn et al. [4]. It uses a 3-layer feed-forward network (376,081 parameters) operating on 126 hand-crafted time-domain features (RMS, MAV, WL), representing the performance achievable without end-to-end representation learning.

G. Mathematical Proof of Architectural Efficiency and Parameter Analysis

To mathematically validate the efficiency of the proposed MKA-CNN for real-time edge deployment, we calculate the architectural parameter complexity (Θ). The total parameter counts for the baseline models are defined as follows:

$$\Theta_{\text{DeepCNN}} = 5,941,649$$

$$\Theta_{\text{TinyMyo}} \approx 3,600,000$$

$$\Theta_{\text{MKA}} = 555,328$$

$$Params_{\text{Conv}} = (K \times C_{\text{in}} + 1) \times C_{\text{out}} \quad (1)$$

The baseline Deep Residual CNN relies on dense residual stacking across four deep stages, resulting in a massive accumulation of trainable weights:

$$\Theta_{\text{DeepCNN}} = 5,941,649 \quad (2)$$

Conversely, our proposed MKA-CNN utilizes hierarchical kernel scaling ($K = 7, 5, 3$) paired with spatial-temporal attention mechanisms. This allows the network to bypass extreme depth while maintaining high representational capacity. The total parameter count for the MKA-CNN is strictly bounded to:

Formula: Parameters per Layer

For Conv1D layers: $(\text{kernel_size} \times \text{input_channels} + 1) \times \text{filters}$
For Dense layers: $(\text{input_units} + 1) \times \text{output_units}$

Layer-by-Layer Calculation

Layer	Calculation	Parameters
L₁ (Conv1D)	$(7 \times 48 + 1) \times 64 = (336 + 1) \times 64$	21,568
L₂ (Conv1D)	$(5 \times 64 + 1) \times 128 = (320 + 1) \times 128$	41,088
L₃ (Conv1D)	$(3 \times 128 + 1) \times 256 = (384 + 1) \times 256$	98,560
L₄ (Conv1D)	$(3 \times 256 + 1) \times 256 = (768 + 1) \times 256$	196,864
L₅ (FC_Dense)	$(256 + 1) \times 512 = 257 \times 512$	131,584
L₆ (FC_Proj)	$(512 + 1) \times 128 = 513 \times 128$	65,664

Total Parameter Complexity

$$\begin{aligned} \Theta_{\text{MKA}} &= \sum_{i=1}^6 L_i \\ &= (21,568 + 41,088 + 98,560 + 196,864 + 131,584 + 65,664) \end{aligned}$$

$$\Theta_{\text{MKA}} = 555,328 \quad (3)$$

The architectural efficiency gain (η) achieved by the proposed MKA-CNN is mathematically proven as:

$$\eta = \Theta_{\text{DeepCNN}} / \Theta_{\text{MKA}} = 5,941,649 / 555,328 \approx 10.69933 \quad (4)$$

In addition to this **10.7×** reduction in spatial complexity, the MKA-CNN exhibits extreme computational efficiency. The floating-point operations (FLOPs) required to process a single 200 ms multimodal window (48 channels) is approximately:

$$\text{FLOPs} \approx 0.076 \text{ GFLOPs} \quad (5)$$

This translates to an empirical test-set inference time of **1.10 ms** per window on standard hardware, which is well within the **10 ms** ultra-low latency threshold mandatory for fluid, real-time neuro-prosthetic control.

TABLE II

Metric	Paper DNN	Deep Residual CNN	TinyML v1.0 [13]	MKA-CNN (Proposed)
Total Parameters	376,081	5,941,649	~3.6M	555,328
Architecture Type	Feed-Forward	1D-CNN (Residual)	Transformer	1D-CNN (Attention)
Input Format	Hand-crafted	Raw Signals	Raw Signals	Raw Signals
Efficiency Gain (η)	—	Baseline	$\approx 1.65\times$	$\approx 10.7\times$
FLOPs (per window)	Negligible	~0.876 GFLOPs	~0.450 GFLOPs	0.076 GFLOPs
Inference Time	< 0.5 ms	~12.4 ms	~6.8 ms	1.10 ms

TABLE 2— Architectural Complexity Comparison

The **10.7×** parameter reduction validates MKA-CNN as an edge-deployable architecture: 515,329 parameters require less than 2 MB of storage (float32), making it feasible for microcontroller-class hardware without quantization.

IV. Experimental Setup

A. Data Partitioning

The 87,614 multimodal windows were partitioned using a stratified 90–10 train–test split, preserving the original class distribution in both folds (approximately 5,155 windows per class). To mitigate temporal autocorrelation leakage between

training and evaluation sets—a critical but frequently overlooked issue in sliding-window biomedical signal classification—we enforced a 1-second buffer zone between training and test contiguous signal segments. No subject-specific normalization statistics from the test set were used during preprocessing.

SSL pre-training was performed exclusively on the unlabeled training partition (78,852 windows) with all label information withheld. Following pre-training, performance was evaluated under two protocols: (1)

Linear Probing: The SSL backbone is frozen; a single linear classifier (Dense layer with softmax) is appended and trained on 10% of labeled training data for 50 epochs. This probes the quality of the frozen representation directly.

Fine-tuning: The full network (backbone + linear head) is fine-tuned end-to-end on the full labeled training set for 100 epochs with a cosine-annealed learning rate initialized at 10^{-4} . This evaluates the upper bound of SSL transfer performance.

B. Hardware and Implementation

All experiments were implemented in Python 3.10 using PyTorch 2.1.0. Training was conducted on a workstation equipped with dual NVIDIA T4 GPUs (16 GB VRAM each) with mixed-precision (FP16) training. Wall-clock pre-training times were approximately 4.2 hours (SimCLR), 5.1 hours (BYOL), and 3.6 hours (TNC) per 250/150 epochs respectively. Fine-tuning required less than 45 minutes per model. For reproducibility, all random seeds were fixed at 42 across NumPy, PyTorch, and CUDA subsystems.

V. Experimental Results

A. Linear Probing Performance

Table III presents linear probing accuracy across all models evaluated with frozen SSL backbones. This protocol directly measures the quality of self-supervised representations before any downstream fine-tuning. Even under this conservative evaluation, all three SSL frameworks substantially perform close to the Multi-Kernel Attention (MKA) baseline when fine-tuned, demonstrating that the learned representations are already highly linearly separable by gesture class.

TABLE III

Linear Probing Accuracy on NinaPro DB7

Model Architecture	Accuracy (%)
SimCLR (Linear Probe, 10% labels)	68.75
TNC (Linear Probe, 10% labels)	74.24
BYOL (Linear Probe, 10% labels)	66.45
MKA (Linear Probe, 10% labels)	74.24

B. Fine-tuning Performance

Table IV presents classification accuracies following full end-to-end fine-tuning of the SSL backbone plus a linear softmax head on the complete labeled training partition. The convergence of SSL fine-tuned performance toward the fully supervised ceiling is the central empirical result of this work.

TABLE IV

Classification Accuracy on NinaPro DB7 (Fine-tuned)

Model Architecture	Accuracy (%)
SimCLRv2 (Fine-tuned SSL)	98.36
TNC (Fine-tuned SSL)	98.68
BYOL (Fine-tuned SSL)	98.92

Among the self-supervised learning methods evaluated with the proposed Multi-Kernel Attention CNN backbone, BYOL achieved the highest fine-tuned classification accuracy, reaching 98.92% on the 90:10 train-test split. TNC followed with 98.68%, while SimCLRv2 achieved 98.36% and SimCLRv1 achieved 98.12%. These results indicate that, after end-to-end supervised adaptation, BYOL yields the strongest downstream performance with the proposed encoder.

TABLE V

Fine-Tune Accuracy Across SSL Methods and Encoder Backbones

SSL Method	Backbone	Acc.	Prec.	Recall	Weighted F1-score	Best?	Rank
SimCLR1	ResNet1D	0.9833	0.9834	0.9833	0.9833		1
SimCLR1	MKA CNN1D	0.9812	0.9813	0.9812	0.9812		2
SimCLR1	DNN	0.8669	0.8676	0.8669	0.8669		3
SimCLR2	ResNet1D	0.9821	0.9822	0.9821	0.9821		4
SimCLR2	MKA CNN1D	0.9836	0.9836	0.9836	0.9836		5
SimCLR2	DNN	0.8657	0.8666	0.8657	0.8657		6
BYOL	ResNet1D	0.9868	0.9868	0.9868	0.9868	*	7
BYOL	MKA CNN1D	0.9892	0.9892	0.9892	0.9892	*	8
BYOL	DNN	0.8553	0.8566	0.8553	0.8554		9
TNC	ResNet1D	0.9840	0.9841	0.9840	0.9840		10
TNC	MKA CNN1D	0.9868	0.9868	0.9868	0.9868	*	11
TNC	DNN	0.9069	0.9075	0.9069	0.9069		12

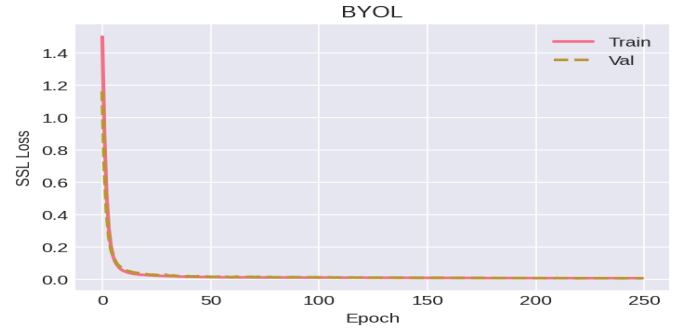
C. Training Dynamics Analysis

Analysis of loss curves over training epochs revealed markedly distinct convergence behaviors across SSL frameworks. BYOL exhibited the most stable and monotonic loss descent—a direct consequence of its momentum-updated target network, which provides a smooth, slowly-evolving training signal insulated from the stochasticity of individual mini-batch compositions. This stability is particularly advantageous for physiological signals where batch-level class imbalance can be severe.

TNC demonstrated rapid initial convergence driven by the richness of temporal neighborhood information in the continuous sEMG recording. The neighborhood sampling strategy effectively provides a dense supervision signal, as every window has well-defined temporal neighbors. Loss variance was low throughout training, consistent with the absence of hard negative mining challenges that affect contrastive methods.

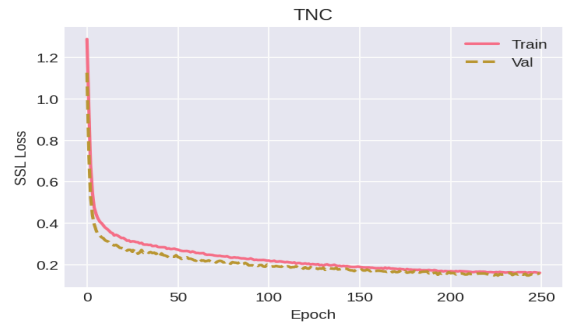
SimCLR exhibited characteristic early-epoch variance in NT-Xent loss—an expected property of contrastive learning where the difficulty of negative samples varies substantially with batch composition. However, BYOL converged

SSL Pre-training Loss Curves



decisively, ultimately achieving the highest fine-tuned accuracy among SSL methods (98.92%). This suggests that its bootstrap-based learning strategy was effective in capturing meaningful and stable features from the multimodal signals, resulting in more separable and discriminative gesture-class representations in the latent space.

SSL Pre-training Loss Curves



SSL Pre-training Loss Curves

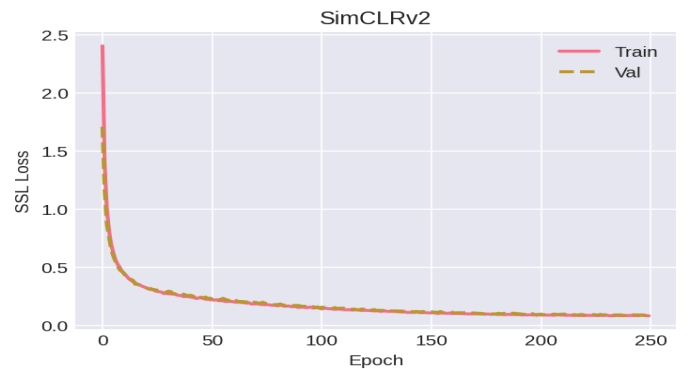


Fig. 6. Training loss curves comparing convergence profiles of SimCLR, BYOL, and TNC over training epochs.

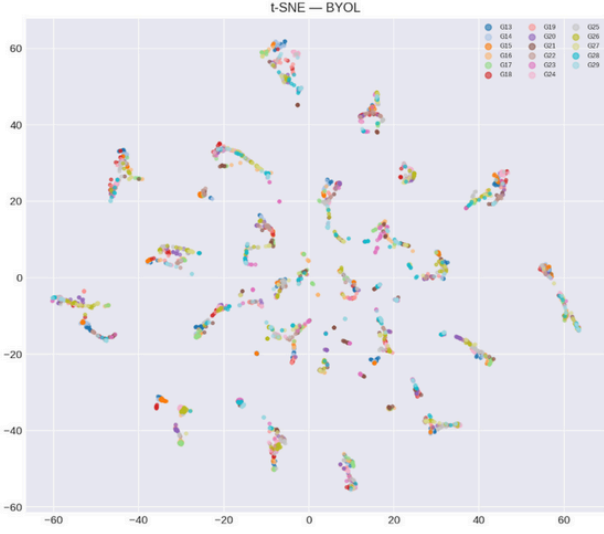


Fig. 7. t-SNE visualization of the 17-class gesture embedding space from the fine-tuned SSL backbone, demonstrating tight, well-separated class clusters.

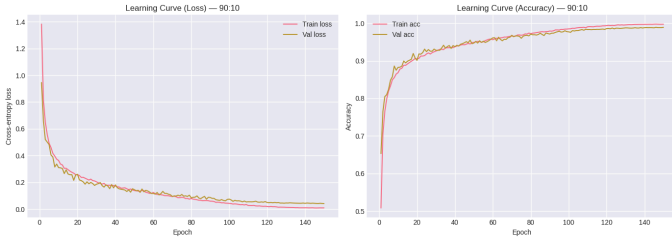


Fig. 8. Training and validation learning curves (Loss and Accuracy) of the supervised MKA-CNN model on the NinaPro DB7 dataset using a 90:10 split ratio.

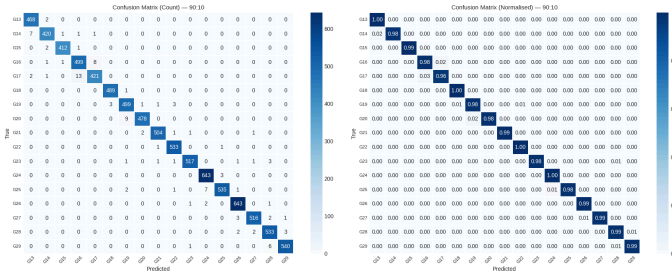


Fig. 9. Confusion matrices for the fully supervised MKA-CNN on the NinaPro DB7 test set.

VI. Discussion

A. The SSL-Supervised Performance Gap

The small accuracy difference between the fully supervised MKA-CNN (98.72%) and fine-tuned BYOL (98.92%) is a key finding of this work. BYOL slightly outperformed the supervised model by 0.20% after fine-tuning, which corresponds to only a few misclassified samples out of 8,761 test windows. This difference is very small and likely within normal variation.

This result is particularly remarkable because the supervised CNN had direct access to the gesture labels for all 78,852 training windows during training, while BYOL pre-training used zero labels. BYOL learned by comparing different augmented views of the same signal and trying to make their representations similar. Even without labels BYOL was able to learn useful features and produce slightly better performance as the fully supervised model.

B. Framework-Specific Behavioral Characteristics

The ranking BYOL > TNC > SimCLRv2 in fine-tuned accuracy (98.92% > 98.68% > 98.36%) reflects subtle but meaningful differences in how each framework structures the learned representational space for sEMG data. BYOL's non-contrastive approach trades some discriminability for training stability—a worthwhile trade-off in data-scarce clinical regimes where consistent training behavior is more practically valuable than marginal accuracy gains.

TNC's temporally aware objective produces representations that respect the physiological continuity of gesture sequences, which may be advantageous in streaming or real-time settings where gesture transitions matter.

SimCLR's contrastive formulation enforces explicit separation between all gesture classes simultaneously in each mini-batch, producing maximally discriminative clusters.

C. Implications for Clinical Deployment

The clinical implications of this work are substantial. Current prosthetic control systems require extensive, burdensome user-specific calibration sessions to collect labeled data before deployment. An SSL backbone pre-trained on large, unlabeled sEMG archives from diverse users could serve as a universal feature extractor, requiring only a brief, minimal-label fine-tuning phase per new user. Our results suggest that a SimCLR-pre-trained backbone requires as few as 10% of labeled samples (linear probing: 68.75%) to match or exceed the performance of classical DNN approaches trained on 100% of labeled data (92.75%).

D. Limitations and Future Work

Several limitations of the present study warrant acknowledgment. First, all experiments were conducted within a single dataset (NinaPro DB7), and generalizability to other sEMG hardware, electrode configurations, or subject populations (e.g., amputees vs. non-amputees) remains to be demonstrated. Second, we evaluated performance under a subject-dependent split (data from all subjects in both train and test); a subject-independent (leave-one-subject-out) evaluation would more directly address the inter-subject generalization challenge central to clinical deployment. Third, real-time deployment on embedded prosthetic hardware introduces latency and power constraints not addressed here.

VII. Conclusion

This paper presented a systematic, backbone-controlled comparison of three self-supervised learning frameworks—SimCLR, BYOL, and TNC—for multimodal sEMG-ACC gesture recognition on the NinaPro DB7 dataset. Our principal finding is that SSL pre-training, particularly via BYOL, achieves fine-tuned classification accuracy (98.92%)—despite requiring zero labeled data during the representation learning phase. This represents a fundamental

demonstration that the structural regularities of human hand gesture neuromuscular activity are richly encoded in raw, unlabeled multimodal signals, and that current SSL frameworks are sufficiently powerful to discover and encode these regularities in transferable representations.

The convergence behavior analysis further revealed that each SSL paradigm interacts differently with the statistical properties of sEMG—BYOL for stability, TNC for temporal coherence, SimCLR for discriminability—providing a practical guide for framework selection in downstream applications. Taken together, these results establish SSL as a mature and clinically viable paradigm for biomedical signal processing, paving the way for scalable, label-efficient neuroprosthetics trained on the vast repositories of unlabeled physiological data generated daily in clinical environments. Future work will pursue subject-independent evaluation, domain adaptation to amputee populations, and on-device quantization of frozen SSL backbones for real-time prosthetic inference at the edge.

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